

Morning Paper Report - Friday, June 05, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Task-Optimized Artificial Neural Networks Align with Human Brain Activity in a Visual Working Memory Task

Pravish Sainath, Guillaume Lajoie, Lune Bellec
2025

Source: Crossref; matched terms: visual working memory, working memory, neural network, neural networks, recurrent neural network

Link: https://doi.org/10.31234/osf.io/7g9ej_v1

Goal: Brain encoding using Artificial Neural Networks (ANNs) has recently emerged as a powerful tool to uncover which features of complex stimuli drive brain activity in a fully data-driven manner

Method: While previous research has primarily focused on perception (vision, audition), this study explores the application of recurrent neural networks (RNNs) to model visual working memory through a 2-back task

Results: Results demonstrate that ANNs can perform the 2-back task with high accuracy, and sparse attention mechanisms were beneficial for a fixed network capacity

Conclusion: The artificial representations of images effectively encoded individual brain activity during 2-back trials. High encoding accuracy was consistently observed in regions such as the dorsal and visual dorsal streams, and frontal cortices.

2. Target Fidelity, but not Inducer Fidelity, Modulates Repulsive Serial Dependence of Visual Working Memory Representations

Bugay Yildirim, Aysecan Boduroglu
2025

Source: Crossref; matched terms: visual working memory, working memory representation, working memory, fidelity, attentional blink

Link: <https://doi.org/10.21203/rs.3.rs-7835332/v1>

Goal: Abstract Visual representations are open to systematic functional biases

Method: In visual working memory, representations of serially presented stimuli can either attract or repel each other. Such serial dependence biases are thought to reflect adaptive Bayesian processes yielding optimal perceptual decisions under uncertainty.

Results: Our findings show adaptive error reduction in working memory, beyond Bayesian optimality.

Conclusion: This perspective suggests that serial dependence should adapt to the degree of uncertainty in both the current target and the inducer (previous target) representations.

3. Predicting visual working memory errors with a priori measures of representational geometry from human and artificial systems

Authors unavailable
2026

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory, modeling, neural network, neural networks

Link: https://osf.io/preprints/psyarxiv/k3xu6_v1/

Goal: A longstanding question in cognitive science is whether the structure of mental representations — deriving from perceptual and semantic knowledge — can be used to generate a priori quantitative predictions of the specific errors people make in memory

Method: Using both human similarity judgments and an ensemble of modern pretrained deep neural networks trained for general visual tasks, with no memory-task or face-identification training, we measure the representational geometry of face stimuli and simple visual features (e.g., color, shape, orientation, location) independently of any memory experiment

Results: Because the same geometric measurement underlies predictions across different aspects of behavior, the results suggest that representational geometry, measured with modern tools, provides a direct and falsifiable a priori link between internal representations and the errors people make in memory.

Conclusion: Because the same geometric measurement underlies predictions across different aspects of behavior, the results suggest that representational geometry, measured with modern tools, provides a direct and falsifiable a priori link between internal representations and the errors people make in memory.

4. Critically synchronized brain waves form an effective, robust and flexible basis for human memory and learning

V. Galinsky, L. Frank
Scientific Reports, 2023

Source: Semantic Scholar; matched terms: flexibility, computational model, computational models, neural network, neural networks, artificial intelligence

Link: <https://www.semanticscholar.org/paper/9ce7d2d9ffa112ca7cbaf5c3d5731e460a510d5f>

Goal: The effectiveness, robustness, and flexibility of memory and learning constitute the very essence of human natural intelligence, cognition, and consciousness

Method: This lack of a solid theoretical framework has implications not only for our understanding of how the brain works, but also for wide range of computational models developed from the standard orthodox view of brain neuronal organization and brain network derived functioning based on the Hodgkin–Huxley ad-hoc circuit analogies that have produced a multitude of Artificial, Recurrent, Convolution, Spiking, etc....

Results: These results have implications for both our understanding of brain function and for the wide range of AI/ML applications.

Conclusion: We demonstrate that the learning inspired by these critically synchronized brain waves is shallow, yet its timing and accuracy outperforms deep ARCSe counterparts on standard test datasets These results have implications for both our understanding of brain function and for the wide range of AI/ML applications.

5. Biologically plausible gated recurrent neural networks for working memory and learning-to-learn

Alexandra R. van den Berg, Pieter R. Roelfsema, Sander M. Bohte
2023

Source: Crossref; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: <https://doi.org/10.1101/2023.07.06.547911>

Goal: Abstract The acquisition of knowledge does not occur in isolation; rather, learning experiences in the same or similar domains amalgamate

Method: While memory models that exclusively use local information for their weight updates have been developed, they have limited capacity to integrate information over long timespans and therefore cannot easily learn-to-learn

Results: Moreover, we show that RECOLLECT can learn-to-learn an effective policy on a reversal bandit task

Conclusion: While memory models that exclusively use local information for their weight updates have been developed, they have limited capacity to integrate information over long timespans and therefore cannot easily learn-to-learn